

Andrei-Ciprian Barbuta

SOFTWARE ENGINEER

Bucharest, Romania

☎ (+40)786453379 | ✉ ciprianbarbuta@gmail.com | 🏠 andreiciprian.com | 📷 ciprix19 | 🌐 ciprix19

Summary

Software engineer with experience in full-stack development, cloud platforms, and enterprise API technologies. My background includes building web applications and supporting enterprise customers using Apigee X and Apigee Hybrid. I have a strong interest in building reliable and maintainable software, alongside cloud infrastructure and automation.

Education

Babes-Bolyai Faculty of Mathematics and Computer Science | *Cluj-Napoca, Romania*

MASTER'S STUDIES IN SOFTWARE ENGINEERING | *October 2023 - April 2024*

- Relevant coursework: Agile Methodologies, Programming Paradigms

BACHELOR'S DEGREE IN COMPUTER SCIENCE | *October 2020 - July 2023*

- Relevant coursework: Data Structures and Algorithms, OOP, Networking, Web Development, Machine Learning
- Thesis: **Tuner and Tab Writer for Guitar Using Note Detection Algorithms**

Work Experience

Arkoudsoft | *Cluj Napoca, Romania*

SOFTWARE ENGINEER | *April 2026 - July 2026*

- Contributed to the **design and development** of a cross-platform **web and mobile festival application** using Tauri, React, and TypeScript.
- Developed frontend features including **interactive festival schedules, swipe-based band preference functionality, offline data caching, notifications, and a festival map editor**, focusing on responsive mobile behaviour.
- Developed and integrated **REST APIs** using Express.js, connecting application features with **PostgreSQL / Supabase** and designed tables and relationships.
- Implemented **offline-first functionality** using IndexedDB, including local caching and synchronization of user changes between offline and online states.
- Collaborated with team members using **Git-based workflows**, resolving merge conflicts, reviewing code, debugging issues, and contributing to architectural and implementation decisions.
- Technologies: **React, TypeScript, Node.js / Express.js, IndexedDB, Tauri, PostgreSQL, Supabase, Docker, Git, HTML5, CSS3.**

Cognizant Softvision | *Bucharest, Romania*

CLOUD ENGINEER L1 - APIGEE & GCP | *May 2024 - September 2025*

- Provided advanced technical support for **Apigee X and Apigee Hybrid**, troubleshooting production and non-production environments for enterprise clients.
- Resolved over **200 customer-reported cases** involving API proxies, environments, deployments, traffic routing, authentication mechanisms, runtime components, and platform configuration.
- Monitored cloud-native platforms using **Google Cloud Logging** and metrics dashboards to track API traffic, service health, and latency across distributed environments.
- Troubleshoot Apigee Hybrid installation and upgrade issues, identifying problems across **Kubernetes clusters**, runtime components, and configuration dependencies.
- Handled role-based access and permission issues, troubleshooting **IAM, service accounts, and policy configurations** affecting API access and platform operations.
- Diagnosed and resolved networking-related issues, including **VPC connectivity, firewall rules, ingress configurations, TLS certificates, and load balancer behavior**, and hybrid communication paths between on-prem and cloud environments.
- Collaborated closely with engineering teams to **reproduce, analyze and escalate bugs and service outages**, ensuring efficient resolution.
- Maintained and continuously improved internal documentation, enabling faster incident resolution and strengthening team knowledge sharing.
- Participated in **high-priority incident management**, ensuring compliance with SLA commitments and minimizing customer impact.
- Technologies used: **Apigee X / Apigee Hybrid, Google Cloud Platform** — IAM, Cloud Logging, Cloud Monitoring, Kubernetes, **Google Cloud Networking** — VPCs, HTTP(S) Load Balancers, **Terraform Modules Apigee.**

Accenture Industrial Software Solution SA | *Cluj-Napoca, Romania*

FRONTEND DEVELOPER INTERN | *August 2022 - November 2022*

- Participated in the design and development of a **secure, Angular-based frontend application** for **API key management**.

- Collaborated with cross-functional team members to translate technical requirements into **scalable, user-friendly UI components**.
- Implemented **best practices** in **TypeScript** and **Angular** to ensure **maintainable, modular, and high-performance code**.
- Actively contributed to **code reviews and iterative improvements**, enhancing **application stability** and **user experience**.
- Technologies used: **TypeScript, HTML5/CSS3, Angular, REST APIs, Docker**.

Coding Projects

Personal Website Portfolio

- Developed a responsive **portfolio website** using **React** and **TypeScript**, showcasing projects, technical skills, and professional experience.
- Containerized the application using a **multi-stage Docker build** with **NGINX** and deployed it on **AWS EC2**, configuring **HTTPS**, **DNS**, **security groups**, and a custom domain.
- Developed a serverless visitor-counter API using **API Gateway**, **AWS Lambda**, and **DynamoDB**, with **CORS** configuration, **API throttling**, and **client-side caching**.
- Built a **CI/CD pipeline with GitHub Actions**, automating linting, production builds, Docker image creation, publishing versioned images to **Amazon ECR**, and deployment to EC2 through **AWS Systems Manager (SSM)**.
- Implemented secure **GitHub Actions** → **AWS authentication** using **Github OIDC** and **IAM roles** in order to avoid long-lived AWS credentials.
- Technologies used: **React, TypeScript, Docker, NGINX, Git, GitHub Actions, AWS** - EC2, ECR, Lambda, API Gateway, IAM, SSM, **DynamoDB, Linux, Cloudflare**.

Real-Time Multiplayer Chess Platform

- Built a **real-time, full-stack multiplayer chess platform** with **authoritative server-side game logic** and **WebSocket synchronization** using **Socket.IO**.
- Complete chess ruleset implemented, including **legal move generation**, **turn validation** and **check/checkmate detection**.
- Server-side move validation to **prevent client-side cheating**.
- Designed a **centralized match state** containing the chessboard, players, assigned colors, and legal moves, ensuring **consistent game state across connected clients**.
- Developed interactive frontend logic for piece selection, **legal move highlighting**, **board synchronization**, and **turn enforcement**.
- Implemented **secure authentication** using access tokens and **HTTP-only cookie-based refresh tokens**, including authentication middleware for protected WebSocket connections.
- Technologies: **Express.js, React, TypeScript, Socket.IO, JSON Web Tokens, HTML5, CSS3**.

Note-Wizard - Tuner and Tab Writer for Guitar Using Note Detection Algorithms

- Bachelor's Thesis JavaFX application which combines a **digital guitar tuner and real-time tab writer**, using **FFT (Fast Fourier Transform)** and signal processing for **accurate note detection**.
- Technologies used: **JavaFX, TarsosDSP**.

Skills

- **Languages** - Typescript, Javascript, Java, C++, C#
- **Frontend** - React, Angular, HTML5, CSS3
- **Backend** - Node.js, Express.js
- **Databases** - PostgreSQL, Supabase, DynamoDB
- **Cloud and Devops** - GCP, AWS, Docker, Github Actions, Terraform, Linux, Git
- **APIs** - Apigee X / Hybrid, REST
- **Soft Skills** - Problem Solving, Collaboration, Adaptability, Communication, Empathy

Languages

- **Romanian** - Native
- **English** - C1
- **German** - A2

Extracurricular Activity

Hermes Society Students UBB | *Cluj-Napoca, Romania*

VOLUNTEER | *Oct 2020 - Oct 2021*

- Demonstrated **teamwork, leadership, and organizational skills** while coordinating tasks and assisting peers.

Untold | *Cluj-Napoca, Romania*

VOLUNTEER | *August 2019*

- Assisted in **event coordination, crowd management, and on-site support**, ensuring a smooth and safe experience for attendees.